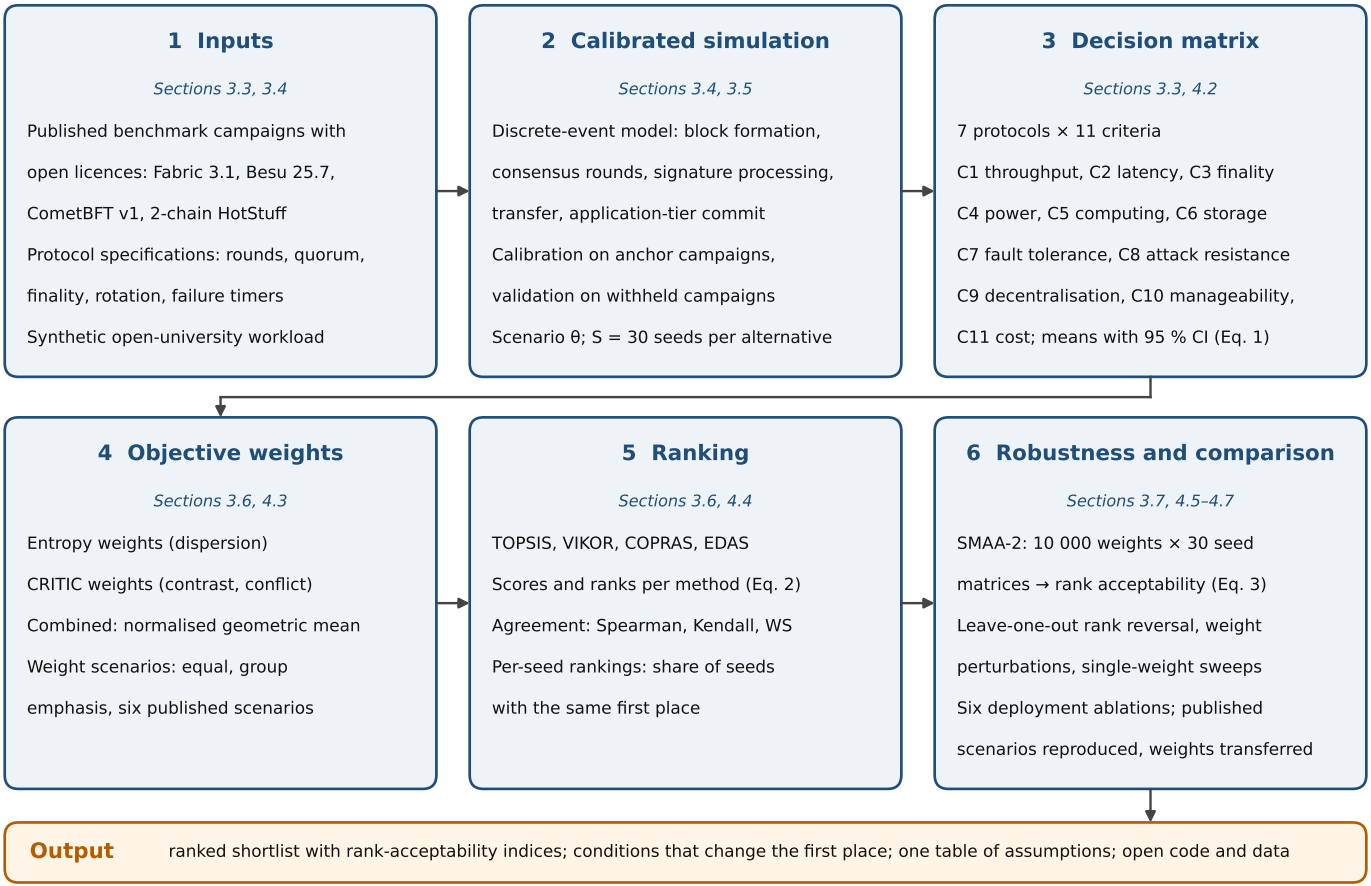
1 Inputs
Sections 3.3, 3.4
Published benchmark campaigns with
open licences: Fabric 3.1, Besu 25.7,
CometBFT v1, 2-chain HotStuff
Protocol specifications: rounds, quorum,
finality, rotation, failure timers
Synthetic open-university workload
2 Calibrated simulation
Sections 3.4, 3.5
Discrete-event model: block formation,
consensus rounds, signature processing,
transfer, application-tier commit
Calibration on anchor campaigns,
validation on withheld campaigns
Scenario θ; S = 30 seeds per alternative
3 Decision matrix
Sections 3.3, 4.2
7 protocols × 11 criteria
C1 throughput, C2 latency, C3 finality
C4 power, C5 computing, C6 storage
C7 fault tolerance, C8 attack resistance
C9 decentralisation, C10 manageability,
C11 cost; means with 95 % CI (Eq. 1)
4 Objective weights
Sections 3.6, 4.3
Entropy weights (dispersion)
CRITIC weights (contrast, conflict)
Combined: normalised geometric mean
Weight scenarios: equal, group
emphasis, six published scenarios
5 Ranking
Sections 3.6, 4.4
TOPSIS, VIKOR, COPRAS, EDAS
Scores and ranks per method (Eq. 2)
Agreement: Spearman, Kendall, WS
Per-seed rankings: share of seeds
with the same first place
6 Robustness and comparison
Sections 3.7, 4.5–4.7
SMAA-2: 10 000 weights × 30 seed
matrices → rank acceptability (Eq. 3)
Leave-one-out rank reversal, weight
perturbations, single-weight sweeps
Six deployment ablations; published
scenarios reproduced, weights transferred
Output
ranked shortlist with rank-acceptability indices; conditions that change the first place; one table of assumptions; open code and data